

EE Simulation Scenarios

Macro and Energy Outcomes

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- Scenarios shown: EE_PDP8, EE_Directive10, EE_Directive10_NoBESS, EE_PDP8_PV_BESS_NoBESS.
- Comparison against Baseline included via explicit deviation plots.
- Horizon in figures: 2026–2050.

How Expert Input Enters the Model

- The Outlook email is not used directly by the model; it is first translated into structured annual inputs.
- Rooftop-solar milestones from the email are consolidated into:
 - `ExcelFiles/Output/RTS_split_assumptions_from_expert_email.csv`
 - `ExcelFiles/Vietnam_EnergyExpert_ScenarioInputs.xlsx`
- The maintenance scripts then read those files and write exogenous scenario paths into the model workbooks:
 - `scripts/maintenance/CreateBaselineFromUserInputFile.m`
 - `scripts/maintenance/CreateEEScenariosFromExpertInputs.m`
- Those workbook paths are what Dynare/MATLAB loads into the model as shocks and state-variable inputs.

How Scenario Inputs Enter the Model

- Source workbook sheets are converted into model scenario sheets by the maintenance pipeline.
- The expert email informs rooftop-solar annual totals and sector splits, which are then propagated into the workbook-driven paths.
- Sector EE savings map to productivity shifters:
 - Industry EE savings → `exo_AI_4_1_2`
 - Services EE savings → `exo_AI_5_1_2`
- EE and RTS investment flows map to adaptation capital stocks:
 - Industry investment → `exo_GA_4_1`
 - Services investment → `exo_GA_5_1`
 - Rooftop solar capacity / generation → `exo_PV_1` and `exo_PVEff_1`
- BESS channels are represented by:
 - PV integration effectiveness → `exo_PVEff_1`
 - BESS annual investment cost stock → `exo_GA_3_1`
- Household RTS investment is accumulated in `exo_PV_1`.

Industrial and Commercial PV: How Demand Is Reduced

Implemented channel for subsectors 4 (industry) and 5 (commercial/services):

- PV generation from the RTS split dataset is converted to coverage shares:

$$\phi_t^s = \min\left(\frac{\text{gen}_{PV,t}^s}{D_t^s}, 0.9999\right), \quad s \in \{\text{ind}, \text{com}\}$$

with demand anchors $D_0^{ind} = 177,550$ GWh and $D_0^{com} = 56,950$ GWh, both grown at 4%/yr.

- Grid-energy demand after behind-the-meter PV is interpreted as:

$$D_t^{grid,s} = D_t^s (1 - \phi_t^s)$$

- The model writes this as an *energy-intermediate productivity gain* (not as direct subtraction from a demand series):

$$\Delta \text{exo_AI}_t^s = \ln\left(\frac{1 - \phi_0^s}{1 - \phi_t^s}\right)$$

and adds it to `exo_AI_4_1_2` (industry) and `exo_AI_5_1_2` (commercial).

- Cost-side capital channel from sector PV capacity is added through `exo_GA_4_1` and `exo_GA_5_1`.

Households and Other Rooftop Segments: Implementation Diagram

RTS split logic used in the baseline pipeline

- If explicit columns exist: use `cap_commercial_MW`, `cap_residential_MW`, `gen_commercial_GWh`.
- Otherwise: split household residual with a configurable share (default 30% to commercial):

$$\text{commercial} = 0.30 \times \text{household}, \quad \text{residential} = 0.70 \times \text{household}$$

Household PV generation path in model variables

RTS residential capacity index $\Rightarrow$ stock-to-investment conversion $\Rightarrow$ `exo_PV_1`

$$\text{exo_PV_1} \Rightarrow I_{PV,r} = (\delta_{PV} \phi_{KPV0} + \text{exo_PV}_r) Y_0$$

$$I_{PV,r} \Rightarrow K_{PV,r} \text{ via } K_{PV,r,t} = (1 - \delta_{PV}) K_{PV,r,t-1} + I_{PV,r,t}$$

$$K_{PV,r} \text{ and } \text{exo_PVEff_1} \Rightarrow Q_{PV,r} = \phi_{PV} K_{PV,r} \exp(\text{exo_PVEff}_r)$$

- For households, PV generation is implemented as home PV production `Q_PV` through `exo_PV_1` and `exo_PVEff_1`.
- For industrial/commercial sectors, the demand-reduction effect is implemented via `exo_AI_4_1_2` and `exo_AI_5_1_2` (plus `exo_GA_4_1/exo_GA_5_1` on the cost side).

PV Transmission Channels: Sector Comparison Schematic

Industry and Commercial (Subsec 4/5)

RTS generation split

$$gen_t^{ind}, gen_t^{com}$$

⇓

Coverage fractions

$$\phi_t^s = \frac{gen_t^s}{D_t^s}, s \in \{ind, com\}$$

⇓

Grid-demand reduction concept

$$D_t^{grid,s} = D_t^s(1 - \phi_t^s)$$

⇓

Implemented model shocks

$$\Delta exo_AI_t^s = \ln\left(\frac{1 - \phi_0^s}{1 - \phi_t^s}\right)$$

Households (Residential RTS)

RTS residential split

$$idx_{PV,t}^{res}$$

⇓

Stock-to-investment conversion

$$idx_{I,t}^{res} \rightarrow exo_PV_1$$

⇓

PV capital accumulation

$$K_{PV,t} = (1 - \delta_{PV})K_{PV,t-1} + I_{PV,t}$$

⇓

Home PV output channel

$$Q_{PV,t} = \phi_{PV} K_{PV,t} \exp(exo_PVEff_1)$$

(displaces household purchased energy)

Translation Equations Used in the Pipeline

- EE savings-to-productivity transform:

$$\Delta AI = \ln \left(\frac{1}{1 - \phi} \right), \quad \phi = \frac{\text{savings_pct}}{100}$$

- Capital-stock accumulation for investment channels:

$$K_A(t) = (1 - \delta_{KA})K_A(t-1) + \frac{\text{Investment}(t)}{GDP_0}$$

with $\delta_{KA} = 0.10$ and $GDP_0 = 430000$ million USD.

- BESS integration gain transform:

$$\Delta_{exo_PVEff}(t) = \ln \left(1 + \frac{\text{PV_Integration_Gain_pct}(t)}{100} \right)$$

- For years beyond expert data, terminal-rate extrapolation extends AI and PVEff paths.
- Rooftop-solar sector splits are reconstructed from the expert email anchors at 2030 and 2050 when explicit commercial and residential series are not present in the workbook.

Scenario Sheet Variables: Explicit Generation (1/2)

Model variable (written)	Source assumption columns	Generation rule in pipeline
exo_AI_4_1_2	Industry_EE_Saving_pct	$\phi_4 = \min(\text{saving}/100, 0.9999)$, $\Delta AI_4 = \ln(1/(1 - \phi_4))$, then $\text{exo_AI_4_1_2} = \text{baseline} + \Delta AI_4$
exo_AI_5_1_2	Services_EE_Saving_pct	$\phi_5 = \min(\text{saving}/100, 0.9999)$, $\Delta AI_5 = \ln(1/(1 - \phi_5))$, then $\text{exo_AI_5_1_2} = \text{baseline} + \Delta AI_5$
exo_GA_4_1	Industry_EE_Investment_USDm + optional	Let $\text{inv}_4 = EE_{ind} + RTS_{ind}$. Accumulate $KA_4(t) = (1 - \delta_{KA})KA_4(t-1) + \text{inv}_4(t)/GDP_0$, then $\text{exo_GA_4_1} = \text{baseline} + KA_4$
exo_GA_5_1	Services_EE_Investment_USDm + optional	Let $\text{inv}_5 = EE_{com} + RTS_{com}$. Accumulate $KA_5(t) = (1 - \delta_{KA})KA_5(t-1) + \text{inv}_5(t)/GDP_0$, then $\text{exo_GA_5_1} = \text{baseline} + KA_5$
exo_PVEff_1	RTS_Services_Investment_USDm PV_Integration_Gain_pct	$\Delta \text{exo_PVEff} = \ln(1 + \text{gain}/100)$, then $\text{exo_PVEff_1} = \text{baseline} + \Delta \text{exo_PVEff}$

If a source column is missing in the expert sheet, the script uses zeros for that component before accumulation.

Scenario Sheet Variables: Explicit Generation (2/2)

Model variable (written)	Source assumption columns	Generation rule in pipeline
exo_GA_3_1	BESS_Annual_Investment_USDbn	Convert bn to million, then accumulate $KA_{ren}(t) = (1 - \delta_{KA})KA_{ren}(t-1) + BESS_{mio}(t)/GDP_0$, and write <code>exo_GA_3_1</code> = baseline + KA_{ren}
exo_PV_1	optional RTS_Household_Investment_USDm	Accumulate household RTS investment stock $KA_{PV}(t) = (1 - \delta_{KA})KA_{PV}(t-1) + RTS_{hh}(t)/GDP_0$, and write <code>exo_PV_1</code> = baseline + KA_{PV}
exo_1AddEE_4_1	configuration flag (not workbook column)	Constant switch written as 1 (additive EE mode in scenario writer)
exo_1AddEE_5_1	configuration flag (not workbook column)	Constant switch written as 1 (additive EE mode in scenario writer)
exo_CapTrade_1	configuration flag (not workbook column)	Constant policy flag written as 1 in EE scenario sheets
Period, Year	baseline time index + aligned expert years	Period is written as sequential index (+1 offset in script), Year is aligned to baseline horizon

NoBESS sheets are generated with the same EE paths but with `exo_PVEff_1`, `exo_GA_3_1`, and `exo_PV_1` reset to baseline values.

Scenario Architecture and Identification

Expert sheet	Full model scenario	Counterfactual
EE_PDP8_reference	EE_PDP8	EE_PDP8_NoBESS
Directive10_RTS_EE	EE_Directive10	EE_Directive10_NoBESS
PDP8_PV_EV_BESS	EE_PDP8_PV_BESS	EE_PDP8_PV_BESS_NoBESS

- Full scenario: EE + RTS + BESS channels active.
- NoBESS scenario: same EE path, but BESS-specific channels revert to baseline.
- Identification logic: Full minus NoBESS isolates the BESS contribution.

How the Main Scenarios Differ in Practice

- EE_PDP8: moderate EE and RTS trajectory (reference policy package).
- EE_Directive10: stronger RTS/EE push with higher integration assumptions.
- EE_Directive10_NoBESS: same Directive10 EE path, but BESS channels switched off.
- EE_PDP8_PV_BESS_NoBESS: accelerated PV package with NoBESS counterfactual settings.

Illustrative 2030 input levels from scenario workbooks:

- RTS capacity (GW): PDP8 revised 25, Directive10 35, PDP8_PV_EV_BESS 60.
- PV integration gain (%): PDP8 revised 2, Directive10 3, PDP8_PV_EV_BESS 5.
- BESS annual investment (USD bn): PDP8 revised 0.374, Directive10 0.903, PDP8_PV_EV_BESS 1.3.

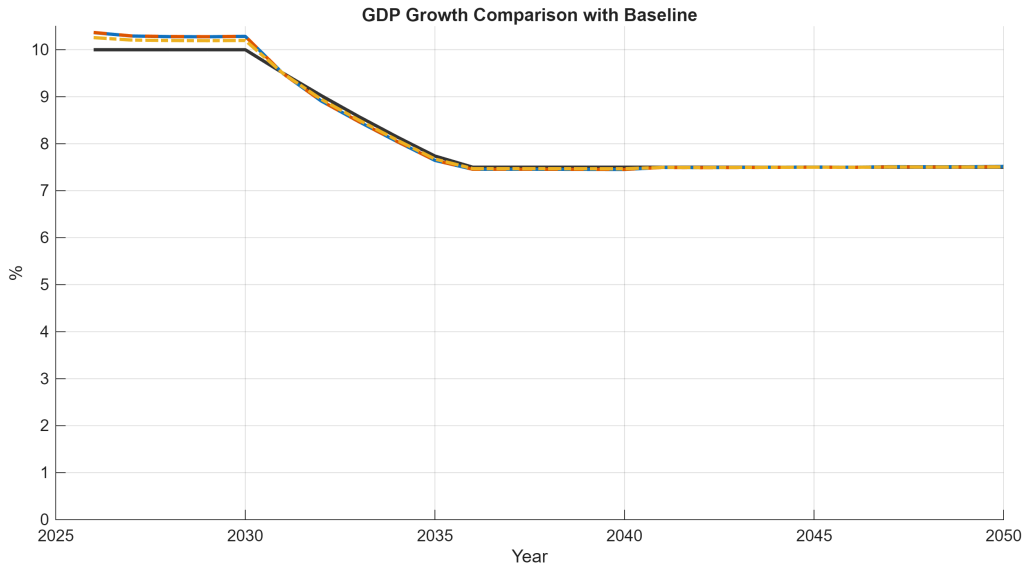
- Directive10 variants show slightly higher GDP levels than EE_PDP8.
- BESS vs NoBESS differences are modest in aggregate macro indicators.
- Energy intensity is lower in Directive10 variants.
- Emissions paths are nearly identical across available EE runs.

Key Numbers (2030, 2040, 2050)

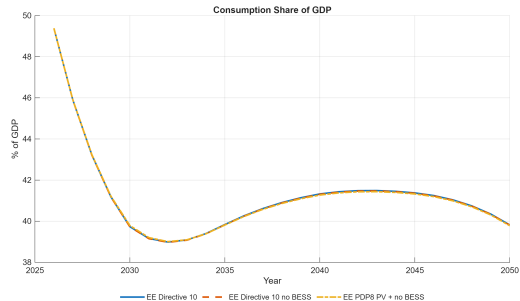
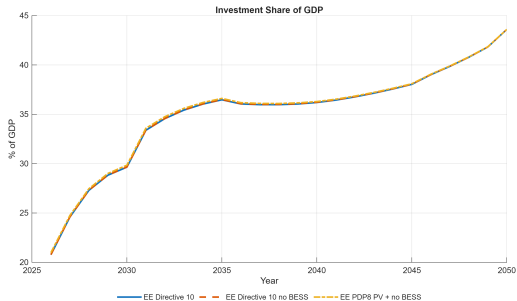
Scenario	2030 GDP growth	2040 GDP growth	2050 GDP growth
EE_PDP8	9.972%	7.503%	7.553%
EE_Directive10	9.963%	7.505%	7.433%
EE_Directive10_NoBESS	9.963%	7.504%	7.455%
EE_PDP8_PV_BESS_NoBESS	9.971%	7.501%	7.508%

- Use the full markdown report for complete table set and definitions.

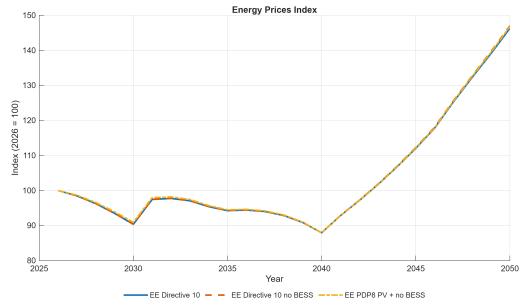
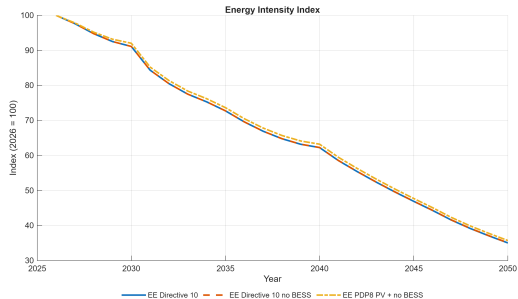
GDP Growth Comparison (Including Baseline)



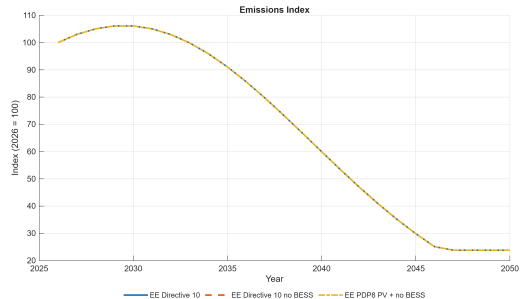
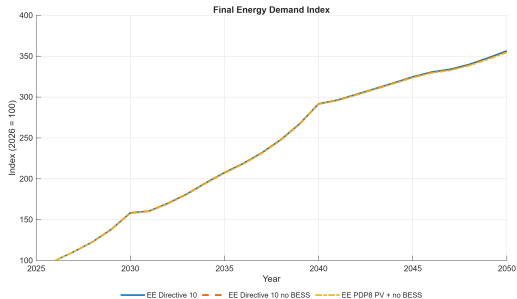
Investment and Consumption Shares



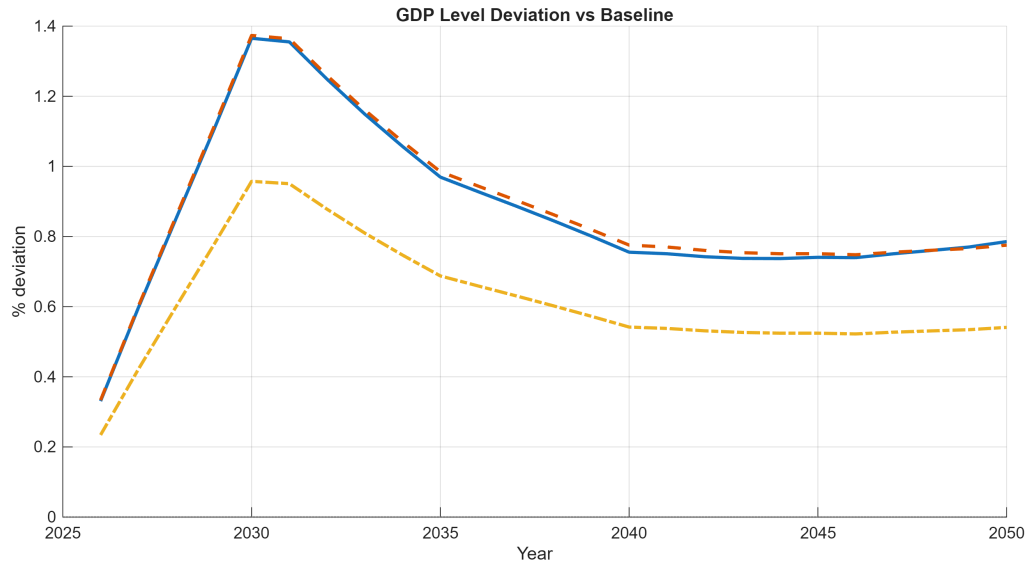
Energy Intensity and Energy Prices



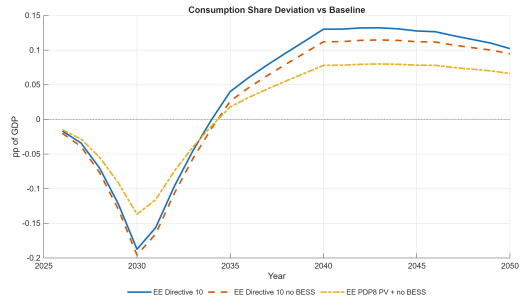
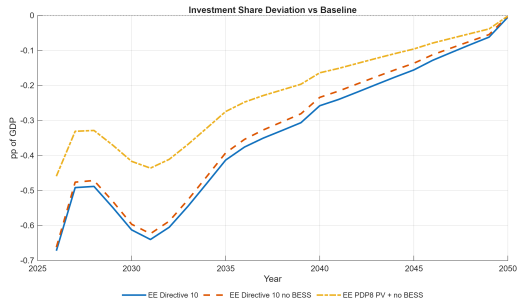
Final Energy Demand and Emissions



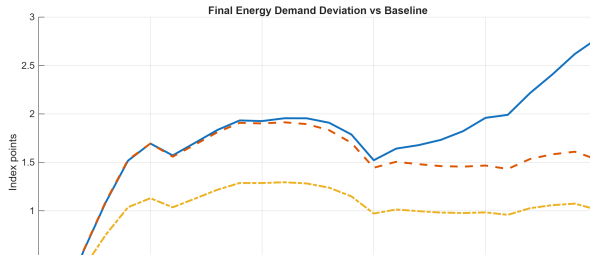
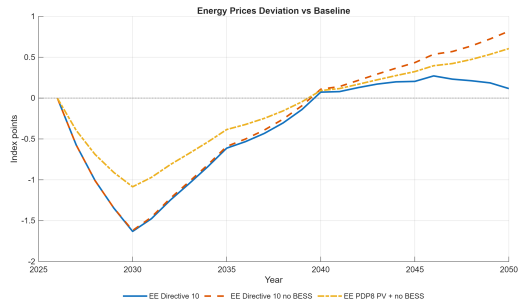
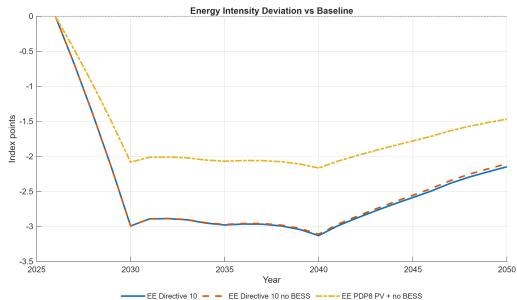
Deviation from Baseline: GDP Level



Deviation from Baseline: Investment and Consumption Shares



Deviation from Baseline: Energy Channels



- EE scenarios modestly improve macro outcomes relative to baseline pathways.
- Directive10 variants tend to deliver stronger efficiency effects.
- Differences between BESS and NoBESS cases are present but moderate in headline macro aggregates.